Causal state-dependent local projections

R practical: participant handout

Name / group: _____

Course date: 19 November 2026

Core practical: 30 minutes. Work in pairs or individually. Record estimates and write short interpretations in the boxes. Optional extensions are for follow-up. No submission deadline or marking scheme is assumed.

Preparation

Install R with the recommended `splines` package. Extract the course folder and set it as the working directory. Open `exercises/participant_template.R`.

```
source("code/R/exercise_setup.R")
```

The setup generates data and defines the estimation functions. It does not run the full horizon loop. All observations are **synthetic**, not DG ECFIN empirical data. The example uses 100 units and 240 dates. The supplied solution script produces numerical answers and a comparison plot.

Model and variables

$$Y_{it} = .6Y_{i,t-1} + (1 - Z_{i,t-1}^2)X_t + c_t + \varepsilon_{it}.$$

The simulation uses independent mean-zero shocks and a newly drawn state each period. Its population response is $g_h(z) = .6^h(1 - z^2)$.

Column	Meaning
`id`, `t`	Unit identifier and shock date
`y`	Outcome at date t
`x`	Common shock at date t
`z`	State already measured before that shock
`ylag`	Outcome at date $t - 1$

Important: Do not lag `z` a second time. A response comparison between two initial states does not identify the effect of intervening on the state itself.

Task 1: sample and timing (7 minutes)

Run:

```
nrow(d)
length(unique(d$id)); length(unique(d$t))
anyDuplicated(d[c("id","t")])
quantile(d$z,c(.05,.5,.95))
all(tapply(d$x,d$t,function(v) length(unique(v))) == 1)
```

1a. Record the number of rows, units and shock dates. Does each unit-date pair occur only once?

1b. Explain why 24,000 panel rows do not give 24,000 independent aggregate shocks.

1c. What is predetermined in this data? Why would using a state measured after the current shock change the interpretation?

Task 2: estimate the impact response (8 minutes)

```
impact <- estimate_horizon(0,Jset=4:10)
impact$J; impact$AIC
grid <- seq(-.8,.8,length.out=161)
curve <- infer_curve(impact,grid)
plot(curve$z,curve$estimate,type="l",
     xlab="Initial state",ylab="Impact response")
lines(grid,1-grid^2,lty=2)
```

2a. Record the selected basis dimension. Compare the fitted curve with the known truth near the centre and near the edges. Attach or save your plot.

Compare a linear interaction using:

```
m <- lm(y ~ x + I(x*z) + ylag,data=d)
glin <- coef(m)["x"]+coef(m)["I(x * z)"]*grid
lines(grid,glin,col="orange")
```

2b. A linear interaction fits $\alpha + \beta z$. Explain why it might report a small β even though the true response varies substantially across states.

2c. Does a good fit prove causal identification in a real dataset? Give one reason.

Task 3: contrasts and uncertainty (8 minutes)

```
contrast <- infer_contrast(impact,zA=0,zB=.75)
contrast
j <- which.min(abs(curve$z))
curve[j,c("estimate","se","point_lower","point_upper",
          "band_lower","band_upper")]
```

3a. Calculate the population impact contrast $g_0(0) - g_0(.75)$ by hand. Record the estimated contrast, its standard error and interval. Does this single interval include the population value?

3b. At the grid point nearest zero, compare the pointwise interval width with the simultaneous-band width. Why do the critical values differ?

3c. Does this band cover all horizons jointly? Does one successful interval establish 95% coverage? Explain both answers.

Task 4: sensitivity and interpretation (7 minutes)

```
fixed <- estimate_horizon(0,Jset=7)
infer_contrast(fixed,zA=0,zB=.75)
```

4a. Compare the fixed-basis result with Task 3. Record whether the point estimate or uncertainty changes materially.

4b. Write a short results paragraph: identify the horizon, shock unit, state comparison and type of interval. Explicitly describe the data as synthetic.

4c. Name one identification check and one inference check you would need before applying the method to your own data.

Hand-in checklist, if requested by the instructor: completed answer sheets, edited R script, one labelled response plot, and the contrast output. Do not submit only screenshots of code.

Optional extensions

A. Horizon 4. Repeat estimation and the contrast at $h = 4$. Compare with the theoretical contrast $.6^4[g_0(0) - g_0(.75)]$.

```
h4 <- estimate_horizon(4,Jset=4:10)
infer_contrast(h4)
h4$dates; h4$n
```

B. HAC bandwidth. Hold the basis dimension fixed and change only the bandwidth. Which part of the result should change?

```
b4 <- estimate_horizon(4,Jset=7,bandwidth=4)
b8 <- estimate_horizon(4,Jset=7,bandwidth=8)
rbind(infer_contrast(b4),infer_contrast(b8))
```

C. Linear benchmark. Fit the linear interaction and compare its response curve with the spline estimate.

```
m <- lm(y ~ x + I(x*z) + ylag,data=d)
names(coef(m))
glin <- coef(m)["x"] + coef(m)["I(x * z)"]*grid
```

D. A policy question. Specify your outcome, shock, pre-shock state and horizon. Explain why the shock could be exogenous and identify the main threat to that argument.

Optional extensions: identification

E. Endogenous state. Generate a state that responds to past shocks:

```
endog <- simulate_panel(state="endogenous")
efit <- estimate_horizon(0,Jset=4:10,data=endog)
infer_contrast(efit)
```

The state update is $Z_{it} = \tanh(.75Z_{i,t-1} + .25X_t + .3v_{it})$. The stored `z` remains $Z_{i,t-1}$. The outcome equation and the independence of future innovations preserve $g_h(z) = .6^h(1 - z^2)$. Explain why an endogenous initial state can be admissible, while conditioning on Z_{it} can change the causal question.

F. Nonlinear shock exposure. Let $f_z(x) = (1 - z^2)x + .25x^3$ with $X \sim N(0, 1)$. Run:

```
nonlinear_shift_demo(z=0,delta=1,cubic=.25)
```

Compare the LP slope times the shift with $E[f_z(X + 1) - f_z(X)]$. Why does a spline in z alone not remove their discrepancy?

Scope of inference. The Gaussian band covers a specified state grid at one horizon under suitable approximation and regularity conditions. AIC selection alone does not guarantee adequate bias control. The practical does not address generated-shock uncertainty or weak instruments.